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PALEOMAGNETISM AND THE ALPINE TECTONICS OF EURASIA IV

JURASSIC, CRETACEOUS AND EOCENE POLE POSITIONS FROM NORTHEASTERN TURKEY

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SUMMARY

In April 1965 and May 1966 several groups of samples were collected by the author in northern and eastern Turkey in view of a study of their magnetic properties.

The characteristic magnetizations of four groups of Eocene and Cretaceous volcanic rocks and sediments had the following directions of magnetization and virtual pole positions (after demagnetization and correction for the tectonical dip):

Eocene	:(Tunceli) $D = 152.5°$, $I = -48°$ and pole 65°N, 66°W.*
Upper Cretaceous-Eocene	:(Gümüshane) $D = 346°$, $I = +40°$ and pole 69.5°N, 98.5°W.
Cretaceous	:(Gümüshane) $D = 153.5°$, $I = -36.5°$ and pole 60.5°N, 81.5°W.
Cretaceous	:(Niskar) $D = 140°$, $I = -36°$ and pole 51°N, 64°W.

For one group of Jurassic sandstones from the Bayburt-Gümüshane region it proved to be very difficult to determine accurately the characteristic direction of magnetization: the samples contained secondary magnetizations, concealing the ancient natural remanent magnetization (N.R.M.). The result is therefore regarded as unreliable. Three groups, varying in age from Permian to Eocene, contained only present-day local geomagnetic field directions and these directions are considered to be secondary.

As the most simple interpretation of previous data and the above mentioned results, the author suggests the hypothesis that Turkey, the Arabian Shield and Africa have acted as one drifting unit since Late Mesozoic times.

INTRODUCTION

The present study forms a part of a paleomagnetic investigation program, carried out by members of the Utrecht University, under the direction of Professors Veldkamp, Rutten and Van Bemmelen, in cooperation

*D = declination; I = inclination

with the American University of Beirut and the Turkish Mineral Research and Exploration Institute (M.T.A.) in Ankara.

A special topic has been the paleomagnetism of the Alpine belts in southeastern Europe and the Middle East (Gregor and Zijderveld, 1964; Raven, 1964; Van Dongen et al., 1967), and the present investigation was made in order to compare Mesozoic results from Turkey with those from Africa, the Arabian Shield and Europe.

Fieldwork and sampling in northern and eastern Turkey was done in April 1965 and May 1966; nine groups of oriented samples were collected, varying in age from Permian to Tertiary. From these the Permian samples from Hazro (east of Diyarbakir), one group of Lower Cretaceous radiolarites and volcanics from Ergani and another group of Eocene volcanics near Bayburt contained only present-day local geomagnetic field directions; the results of five groups of Jurassic (from Bayburt), Cretaceous (from Niksar and Gümüshane) and Eocene age (Gümüshane and Tunceli) are presented here (summarized in Table II), and the results of some Oligocene Miocene volcanics will be published later.

GEOLOGICAL SITUATION

The eastern Pontide region (Fig.2, 3, samples BAY and VTB)

The Pontide chains run from west to east along the Black Sea coast from the Zonguldak-Amasra region to the Trabzon-Rize region (Fig.1). This mountain range lies immediately to the north of a great zone of dextral strike slip faults, together forming the North Anatolian faultzone, which runs from the Sea of Marmora to Lake Van and to Erzurum in eastern Turkey. Arni (1939) and Blumenthal (1964) described the Pontide chains as a mountain range with mainly Cretaceous sediments and volcanics around a few Paleozoic nuclei.

The geology of the *Gümüshane region* (Fig.2) has been described recently by Zankl (1961), who paid special attention to the volcanics of Jurassic and Cretaceous age in a north-south cross-section through the eastern Pontides. Apart from various intrusive complexes, these volcanics consist of flows of basalt, trachy-andesite, dacite, spilite and rhyolite, intercalated between sediments, tuffs and agglomerates. The marine sediments in this sequence (limestones, sandstones and marls) made it possible to attribute most of the volcanics to either the Jurassic or the Early Cretaceous or the Late Cretaceous-Eocene period and to establish their correct tectonical dip.

The *Niksar region* (Fig.3) is a continuation to the west of the Gümüshane region.

Fig.1. Regional map of Turkey, showing the sample areas of Fig.2, 3 and 4.

Fig.2. Map of the Gümüshane–Bayburt region (Pontide chains, northeastern Turkey). The sampling sites are indicated.

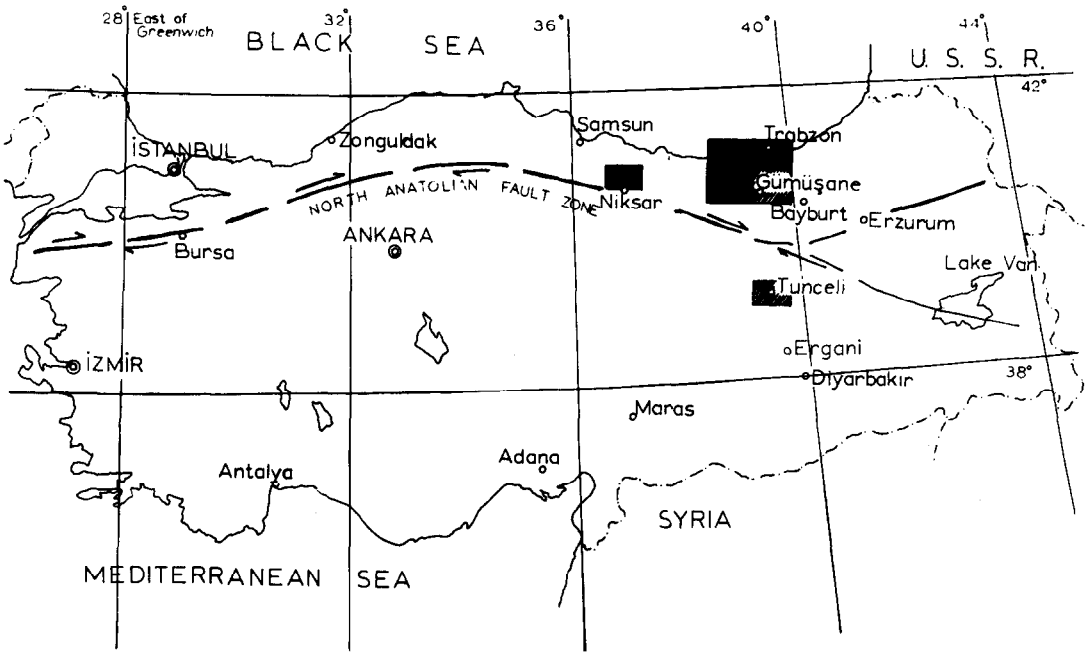

28° East of Greenwich
32°
BLACK SEA
36°
40°
44°
U. S. S. R.
42°
İSTANBUL
Zonguldak
Samsun
Trabzon
Gümüşane
NORTH ANATOLIAN FAULT ZONE
Niksar
Bayburt
Erzurum
Bursa
ANKARA
Lake Van
Tunceli
İZMİR
Ergani
Diyarbakır
38°
Maras
Antalya
Adana
SYRIA
MEDITERRANEAN SEA

Fig.1.

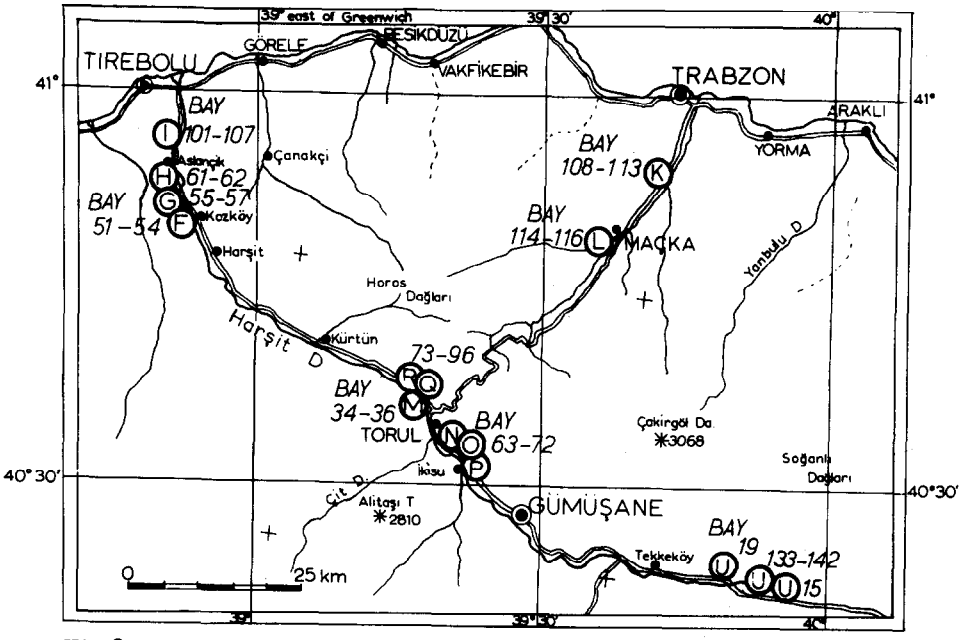

39° east of Greenwich
39°30'
40°
TIREBOLU
GÖRELE
REŞİKDÜZÜ
VAKFİKEBİR
TRABZON
41°
ARAKLI
YÖRMA
BAY
I 101-107
Çanakçı
H 61-62
G 55-57
F
BAY
51-54
Kazköy
Harşit
BAY
108-113 K
BAY
114-116 L
MAÇKA
Yanbulu D.
Horos
Dağları
Harşit D.
Kürtün
73-96
Q
BAY
34-36
M
TORUL
N
O
63-72
P
Çakirgöl Da.
*3068
Soğanlı
Dağları
40°30'
İkisu
Alitaşı T.
*2810
GÜMÜŞANE
BAY
19 133-142
Tekkeköy
U
15
0
25 km

Fig.2.

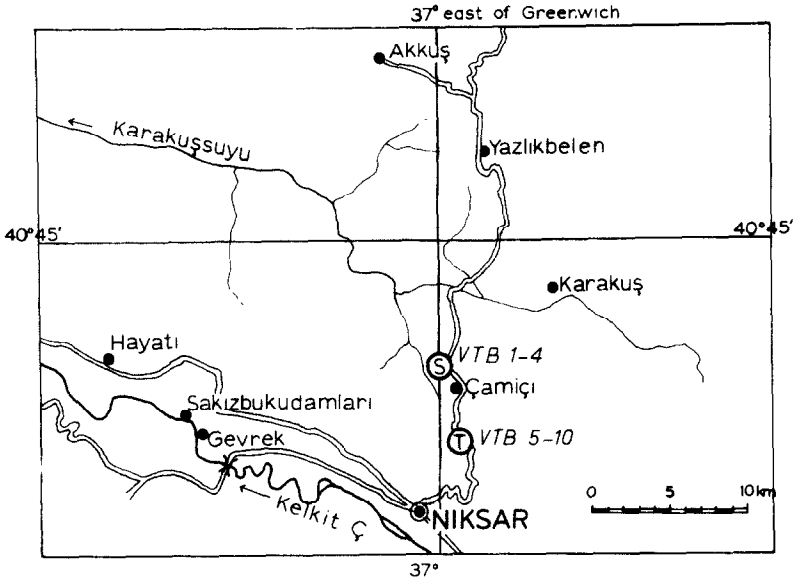


Fig. 3. Map of the Niksar region (Pontide chains, northeastern Turkey). The sampling sites are indicated.

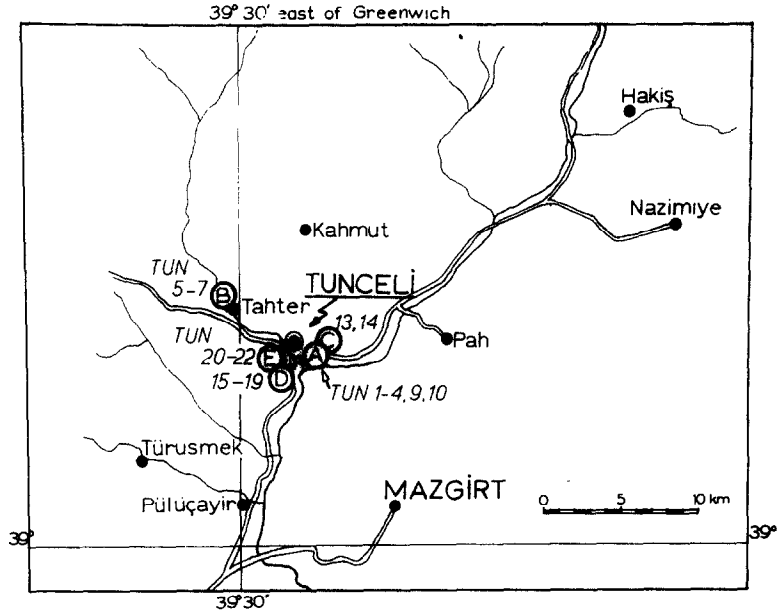


Fig.4. Map of the Tunceli region (Eastern Tauride chains, eastern Turkey). The sampling sites are indicated.

The Tunceli region (Fig.4, samples TUN)

Tunceli lies some 160 km south of Gümüshane, in the eastern Tauride chains (Fig.1).

In the area sampled (Fig.4), the series attributed to the Eocene lies unconformably on Permian metamorphic rocks and is covered unconformably by Miocene conglomerates (Geological Map of Turkey, 1:500,000; sheet Erzurum, 1961). The series consists of an alternation of predominantly red, medium to fine grained, quartz sandstones, siltstones and shales, with some grey-green beds otherwise similar in character. At some places trachyte-flows occur between the sandstone; they are also attributed to the Eocene period (Geological Map of Turkey, 1:500,000; sheet Erzurum, 1961).

METHODS OF RESEARCH

For measurement the samples were sawn to approximately equidimensional shape, in order to minimize shape effects, and embedded in their correct orientation in paraffin cubes with 10 cm edges. Directions and intensities of magnetization were then measured by means of astatic magnetometers in the Paleomagnetic Laboratory of the Utrecht University, by procedures described previously (As, 1960, Van Everdingen, 1960). In order to obtain an analysis of the natural remanent magnetizations (N.R.M.'s) all samples were treated by alternating magnetic fields produced by a.c. in a zero ambient magnetic field, in a progressive partial demagnetization. For an extensive description of the analysis and procedures of the demagnetization see As and Zijderveld (1958) and Zijderveld (1967).

The numerical data thus obtained are summarized in Table I, and Fig.6, 8, 10, 12 and 14 show the data plotted in stereographic projection (by means of the Wulff-net). In Table II the mean characteristic direction for each group and its statistic parameters are listed. The following values of the local geomagnetic variation were taken for correction: Niksar 3.5°E, Gümüshane 4°E and Tunceli, 3.5°E.

MEASUREMENT AND DEMAGNETIZATION

The Jurassic Bayburt samples (Fig.2, 5, 6; samples BAY 15, 19, 134–142)

From different exposures with varying tectonical dip, these samples of red Jurassic sandstone (Geological Map of Turkey, 1:500,000, sheet Trabzon, 1961) and one sample of intercalated trachy-andesites have been collected in order to establish a Jurassic virtual pole position for the Bayburt region. Obviously, however, as can be seen in Fig.5B, these samples are less suitable for paleomagnetic research, in particular for a.c. magnetic field demagnetization treatment.

The N.R.M.'s had intensities of about $30 \cdot 10^{-6}$ e.m.u./cm^3 and consisted of two components: one very large component (approximately 75% of the N.R.M.) with a direction along the present-day geomagnetic field in Turkey, and another component with a direction of magnetization which in most cases was very difficult to determine accurately (Fig.5B). The former magnetization

TABLE I

Mean direction of magnetization for the sites

Sites	Samples	Number of samples	A. N.R.M.			B. After demagnetization			C. After demagnetization and after dip-correction		
			D	I	α_{95}	D	I	α_{95}	D	I	α_{95}
A. Tunceli, Eocene red beds	TUN 1–4, 9, 10	6	138.5°	+1°	39°	147.5°	-18°	9.5°	169.5°	-47.5°	8.5°
B. Tahter, Eocene trachytes	TUN 5–7	3	319.5°	+50.5°	39.5°	212.5°	-48.5°	4°	171.5°	-47°	4°
C. Tunceli, Eocene red beds	TUN 13, 14	2	112°	-0.5°	99°	132.5°	-29°	20.5°	130.5°	-49.5°	9°
D. Tunceli, Eocene red beds	TUN 15–19	5	319.5°	+50°	21°	326°	+40.5°	6°	319°	+45°	5.5°
E. Tunceli, Eocene red beds	TUN 20–22	3	359°	+37°	15°	–	–	–	–	–	–
F. Kozköy, Eocene–Cretaceous limestones	BAY 51–54	4	1.5°	+54°	6°	357°	+51.5°	8°	336.5°	+47°	8°
G. Kozköy, Eocene–Cretaceous rhyolites	BAY 55–57	3	13.5°	+44°	11°	13.5°	+41.5°	10°	4°	+43°	9°
H. Aslançik, Eocene–Cretaceous trachy-andesites	BAY 61, 62	2	358°	+40.5°	11.5°	355°	+37.5°	12°	352.5°	+33.5°	12°
I. Tirebolu, Eocene–Cretaceous trachy-andesites	BAY 101–107	5	352°	+54.5°	11.5°	345°	+52.5°	9.5°	349.5°	+39°	9.5°
K. Trabzon, Eocene–Cretaceous trachy-andesites	BAY 108–113	6	309°	-22°	–	221°	-62.5°	4°	344.5°	+36°	4.5°
L. Maçka, Eocene–Cretaceous rhyodacites	BAY 114–116	3	347°	+69°	15°	347°	+67.5°	7.5°	356°	+38.5°	7.5°
M. Torul, Cretaceous rhyodacites	BAY 34–36	3	163.5°	+38.5°	22°	164°	+34.5°	13°	163°	-40°	13°
N. Ikisu, Cretaceous red sandstones	BAY 63–66	4	142.5°	+31.5°	18°	153°	+10.5°	15°	134.5°	-42.5°	13.5°
O. Ikisu, Cretaceous red sandstones	BAY 67–69	3	131.5°	+43°	17.5°	142.5°	+26°	20°	139°	-31°	20°
P. Ikisu, Cretaceous red sandstones	BAY 70–72	3	152.5°	+35°	9.5°	155.5°	+29°	6°	150°	-36°	6°
Q. Torul, Cretaceous rhyodacites	BAY 73–84	10	317.5°	+16.5°	14.5°	306.5°	-9°	10°	165.5°	-43.5°	10°

R. Torul, Cretaceous rhyodacites	BAY 89–96	8	306.5°	-17°	3°	302°	-28°	3.5°	166°	-23°	3.5°
S. Çamiçi, Cretaceous basalts	VTB 1–4	4	162°	-37°	25°	166°	-44°	18°	166°	-44°	18°
T. Niksar, Cretaceous basalts	VTB 5–10	5	104°	+11.5°	25°	120°	-23.5°	6.5°	120°	-23.5°	6.5°
U. Tekkeköy, Jurassic trachy-andesite and red sandstones	BAY 15, 19, 134–142	8	54°	+60.5°	32°	155°	-35°	9.5°	146.5°	+3.5°	8°

D = declination; I = inclination; α_{95} = semi-angle of cone of 95% confidence, after Fisher (1953).

TABLE II

Mean direction of magnetization for the groups, after demagnetization[1]

Sites	B. Before dip-correction giving unit weight to sites				C. After dip-correction: giving unit weight to samples					giving unit weight to sites				
	D	I	α_{95}	N	D	I	α_{95}	k	N	D	I	α_{95}	k	N
A–E, Eocene, samples TUN 1–19	154.5°	-37.5°	36°	4	155°	-48°	6°	36	16	152.5°	-48°	16°	33	4
F–L, Eocene–Upper Cretaceous, samples BAY 51–62, 101–116	350°	+46°	>60°	6	349°	+40°	3.5°	67	23	346°	+40°	7°	90	6
M–R, Cretaceous, samples BAY 34–36, 63–96	–	–	>90°	6	157.5°	-36°	5.5°	22	31	153.5°	-36.5°	11°	35	6
S–T, Cretaceous, samples VTB 1–10	140°	-36°	(66°)	2	137°	-34.5°	16°	11	9	140°	-36°	–	8	2
U, Jurassic, samples BAY 15, 19, 134–142	155°	-35°	9.5°	8	146.5°	+3.5°	8°	51	8	–	–	–	–	(1)

[1] D = declination, I = inclination; α_{95} = semi-angle of cone of 95% confidence (after Fisher, 1953); k = precision parameter (after Fisher, 1953); N = number of units.

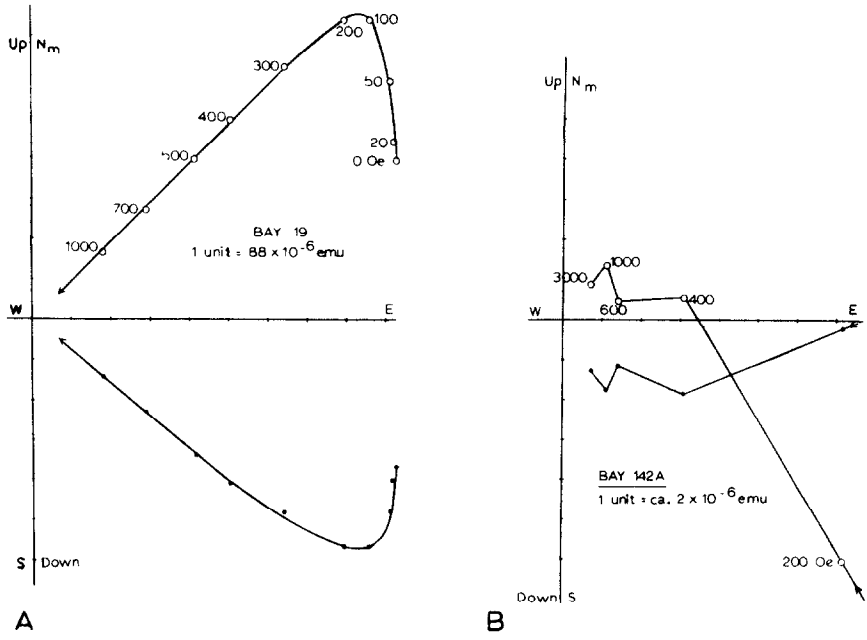


Fig.5. Demagnetization diagrams of two Jurassic samples from Bayburt: A. sample BAY 19, trachy-andesite from Tekkeköy; B. sample BAY 142A, red sandstone from Tekkeköy.

Plotted points represent successive positions - in orthogonal projection - of the end of the magnetic vector during progressive demagnetization. Full symbols represent projection on the horizontal plane, open symbols projections on the east–west vertical plane. Not corrected for the present-day geomagnetic declination (3.5 - 4° E, N_m = magnetic North).

is likely to be secondary and the latter, as it proved to have a similar direction in all samples, is the "characteristic" direction.

The demagnetization of the only sample of the intercalated trachy-andesites (Fig.5A) also reveals a component directed along the present-day local geomagnetic field, which is eliminated in a.c. magnetic fields of about 300 Oe (peak value). In the higher alternating fields, however, another component was eliminated and close examination of Fig.5A reveals that the decrease of the magnetic vector between 400 and 1,000 Oe takes place in a direction that passes approximately through the origin as a straight line, showing that only one component was present in the sample at 400 Oe. The direction of this magnetization is consistent with the characteristic directions obtained from the sandstones, and together they yielded a mean direction of magnetization with D = 146.5°, I = +3.5° and α_{95} = 8° (eight samples, see Fig.6).

Because of the criteria by which one judges the results nowadays, this result is not wholly satisfactory. The author would like to emphasise that this result can be taken as no more than a confirmation of the general conclusions in the next part of this paper.

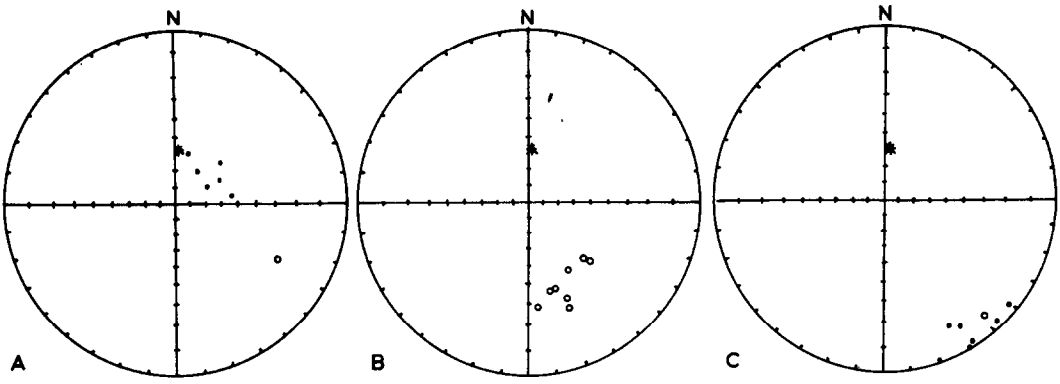


Fig.6. Jurassic trachy-andesite and sandstone samples of Bayburt. Stereographic projections of the directions of magnetization: A. total N.R.M.; B. characteristic directions after demagnetization; C. characteristic directions after demagnetization and after dip-corrections. Full (open) symbols in lower (upper) hemisphere, positive (negative) inclination. The asterisk denotes the present-day geomagnetic field direction in situ.

The Cretaceous Niksar basalts (Fig.7, 8, samples VTB 1–10)

These more or less flatlying basalts were collected from two exposures in the Cretaceous sequence north of Niksar (Geological map of Turkey, 1:500,000, sheet Samsun, 1961). The intensities of their N.R.M.'s varied between $600 \cdot 10^{-6}$ and $4300 \cdot 10^{-6}$ e.m.u./cm^3, their Q-values[1] varied between 0.2 and 3.

The samples behaved more or less identically during the process of demagnetization; a secondary component, which was directed approximately along the recent local geomagnetic field, was eliminated in the a.c. magnetic field trajectory between 0 and 200 Oe, and the higher alternating fields revealed another component (Fig.7). The directions of these "harder" magnetizations appeared to be consistent within the site, and these characteristic directions have been plotted in Fig.8B. As in Fig.5A, we see in Fig.7 that the decrease of the magnetization vector between 400 and 1,200 Oe passes through the origin as a straight line and we may conclude that only one component was present at 400 Oe in the sample.

The samples of the two sites have an average direction of magnetization with $D = 137°$, $I = -34.5°$ and $\alpha_{95} = 16°$ (average of nine samples). Giving unit weight to sites (N=2), the mean direction becomes $D = 140°$ and $I = -36°$.

[1] Q denotes the ratio of the intensities of N.R.M. and induced magnetization ($= J/k.H_Z$, where J is the intensity of the remanent magnetization in e.m.u./cm^3, k is the susceptibility and H_Z is the intensity of the vertical component of the ambient field.) (Utrecht, H_Z = 0.44 Oe.)

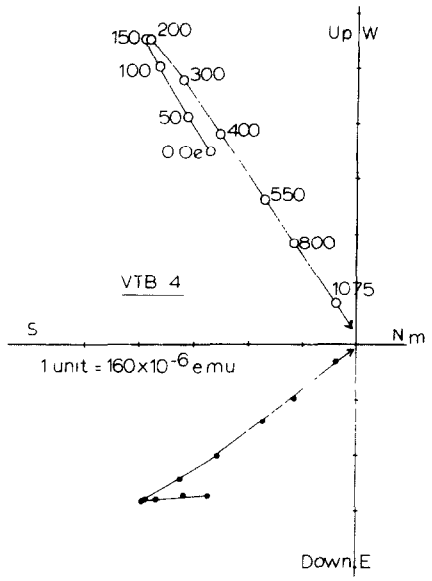


Fig.7. Demagnetization diagram of a Cretaceous basalt-sample from the Niksar region (site S, Çamiçi). Explanation see Fig.5.

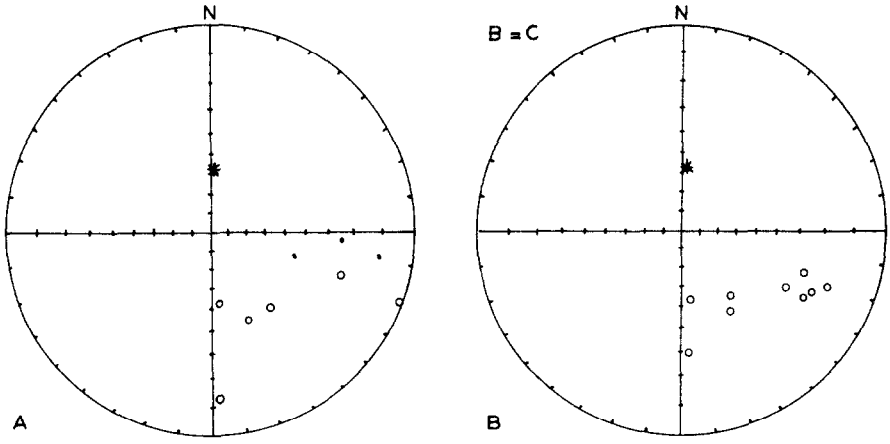


Fig.8. Cretaceous basalts from the Niksar region (sites S and T, Çamiçi). Stereographic projections showing the directions of magnetization: A. total N.R.M.; B. = C. characteristic directions after demagnetization (subhorizontal dip, no correction). Further explanation see Fig.6.

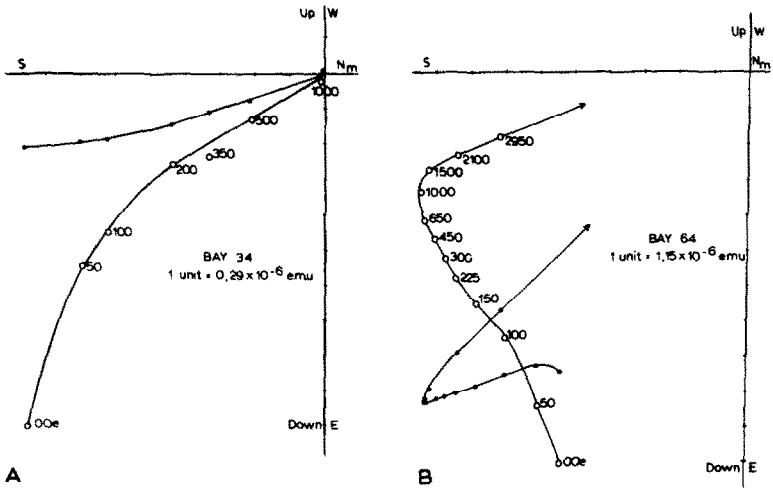


Fig.9. Demagnetization diagrams of two Cretaceous samples from Gümüshane: A. BAY 34, rhyodacite from Torul, site M; B. BAY 64, red sandstone from Ikisu, site N. Explanation see Fig.5.

The Middle or Upper Cretaceous Gümüshane sample group (sites M–R, samples BAY 34–36, 63–96; Fig.9, 10)

The sample group consists of pale red sandstones (sites N - P) and dark grey-green rhyodacites (sites M, Q and R) from exposures with different dips. Their age has been determined after the description by Zankl (1961) and the Geological Map of Turkey, 1:500,000, sheet Trabzon (1961).

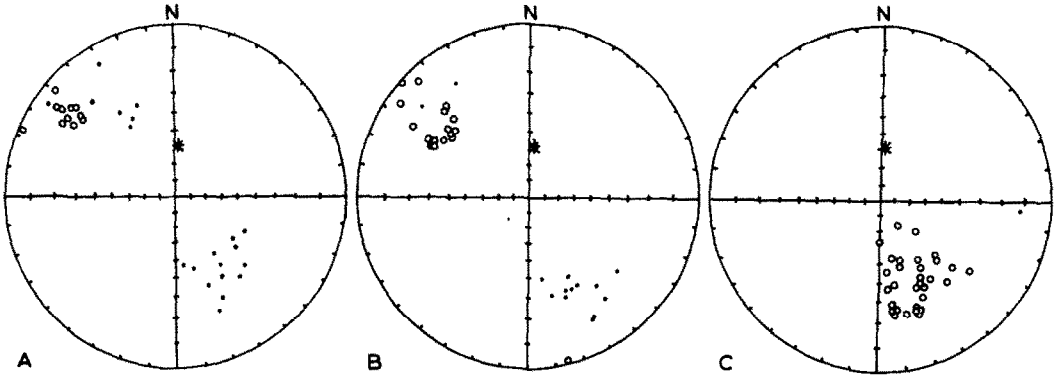


Fig.10. Cretaceous volcanic rocks and sediments from the Gümüshane region. Stereographic projections of the directions of magnetization: A. total N.R.M.; B. characteristic components after demagnetization; C. characteristic components after demagnetization and after dip-correction. Further explanation see Fig.6.

The sandstones had intensities varying between $14 \cdot 10^{-6}$ and $30 \cdot 10^{-6}$ e.m.u./cm^3 and Q-values of about 3. Their softer magnetizations were directed along the present-day local geomagnetic field and were eliminated in the a.c. magnetic field trajectory between 0 and 1,500 Oe (Fig.9B).

The intensities of the volcanic rocks varied between $2 \cdot 10^{-6}$ and $18 \cdot 10^{-6}$ e.m.u./cm^3 and their Q-values were about 0.5. An important part of their N.R.M. had also a direction conforming to the present-day local geomagnetic field; it is assumed that these magnetizations are secondary. These secondary components were easily eliminated in a.c. magnetic fields of up to 300 Oe (Fig.9A).

The harder magnetizations in both sandstones and volcanic rocks appear to have consistent characteristic directions; they are plotted in Fig.10B in stereographic projection, and the average directions of magnetization for the sites are listed in Table I.

It is assumed, though it could not be proved, that the strata of two sites (Q and R) are overturned with a dip of 55° to the southeast. This assumption is justified by the consistent directions of both overturned and moderate dipping strata after unfolding. The characteristic directions of all samples then are grouped around an average direction of magnetization with $D = 157.5°$, $I = -36°$ and $\alpha_{95} = 6°$ (average of 31 samples). Giving unit weight to sites ($N = 6$), the average direction becomes $D = 153.5°$, $I = -36.5°$ and $\alpha_{95} = 11°$. The directions of the characteristic components after unfolding, are plotted in Fig.10C.

The Upper-Cretaceous–Eocene Gümüshane sample group (sites F–L, samples BAY 51–62, 101–116; Fig.11,12)

The group was collected from six sites, of Upper Cretaceous limestone (with *Inoceramus balticus* BOEHM, site F) and of Upper Cretaceous

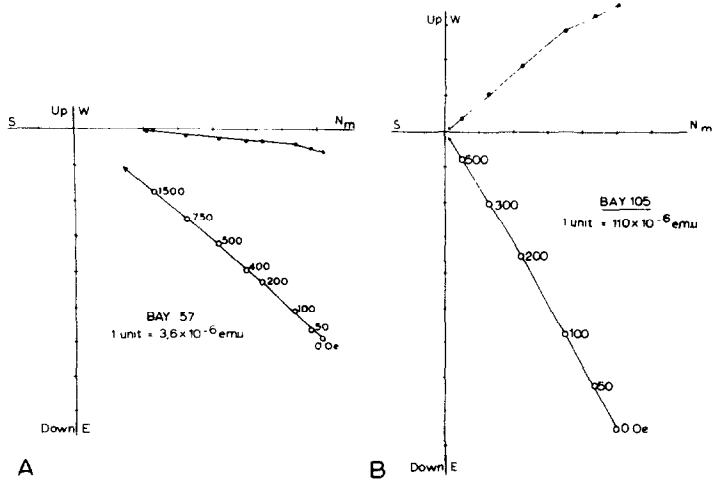


Fig.11. Demagnetization diagrams of two Upper Cretaceous–Eocene samples from Gümüshane: A. BAY 57, rhyolite from Kozköy, site G; B. BAY 105, trachy-andesite from Tirebolu, site I. Explanation see Fig.5.

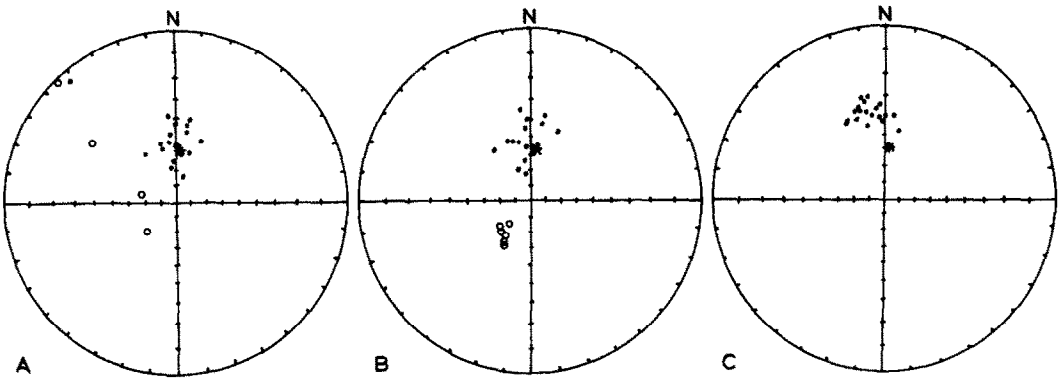


Fig.12. Upper Cretaceous–Eocene volcanic rocks and sediments, from the Gümüshane region. Stereographic projections of the directions of magnetization: A. total N.R.M.; B. characteristic components after demagnetization; C. characteristic components after demagnetization and after dip-correction. Further explanation see Fig.6.

or Lower Eocene volcanic rocks (mainly trachy-andesites and rhyolites), according to Zankl (1961).

The pale-red coloured sandy limestones had intensities varying between $14 \cdot 10^{-6}$ and $28 \cdot 10^{-6}$ e.m.u./cm^3 and remarkably high Q-values varying between 20 and 130.

The intensities of the N.R.M.'s of the volcanic rocks were different for each site and varied between $100 \cdot 10^{-6}$ and $13{,}000 \cdot 10^{-6}$ e.m.u./cm^3 for the trachy-andesites and between $10 \cdot 10^{-6}$ and $33 \cdot 10^{-6}$ e.m.u./cm^3 for the rhyolites (site G).

The sedimentary samples lost their secondary magnetizations in a.c. magnetic fields varying between 500 and 1,000 Oe, the trachy-andesites between 50 and 200 Oe (Fig.11B), and the harder magnetizations, remaining in the samples, are regarded as original. The rhyolites contained only one component which was almost eliminated in 3,000 Oe, with a direction similar to the other samples (Fig.11A). With the exception of site K, which probably had an overturned position with a dip of 30° to the north, most sites had a moderate dip and after unfolding the directions of the samples grouped around an average direction of magnetization with $D = 346°$, $I = +40°$ and $\alpha_{95} = 7°$ giving unit weight to sites ($N = 6$, Fig.12C).

The Eocene Tunceli group (sites A–E, samples TUN 1–22; Fig.13, 14)

The 22 samples of this group were collected from five sites: one site of Eocene trachytes (site B), the other sites of red siltstones, deliberately chosen from beds with different tectonic dips.

The intensity of the N.R.M. of the trachytes varied between $80 \cdot 10^{-6}$ and $150 \cdot 10^{-6}$ e.m.u./cm^3. Their Q-values were about 0.1. Progressive demagnetization of these samples revealed a strong component conforming to the present-day geomagnetic field (Fig.13A). This secondary component was

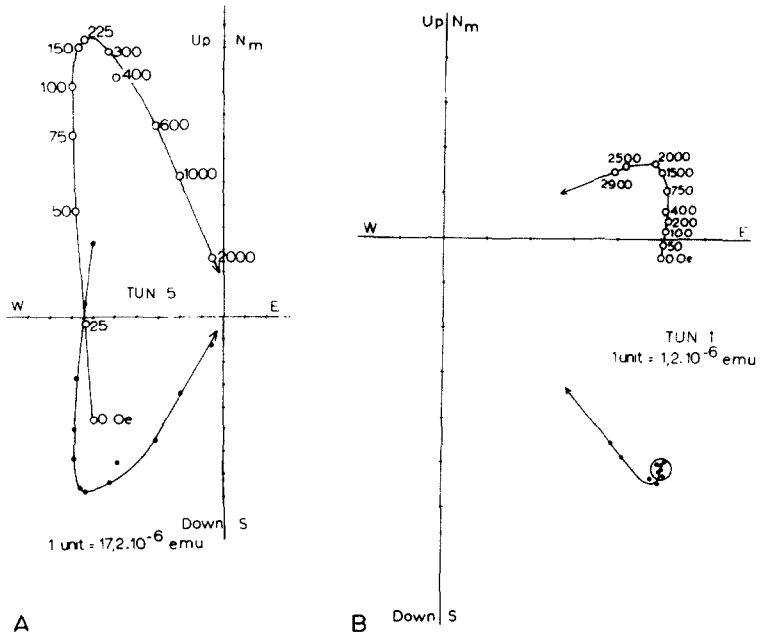


Fig.13. Demagnetization diagrams of two Eocene samples from the Tunceli region: A. TUN 5, trachyte-sample from Tahter, site B; B. TUN 1, red sandstone from Tunceli, site A. Explanation see Fig.5.

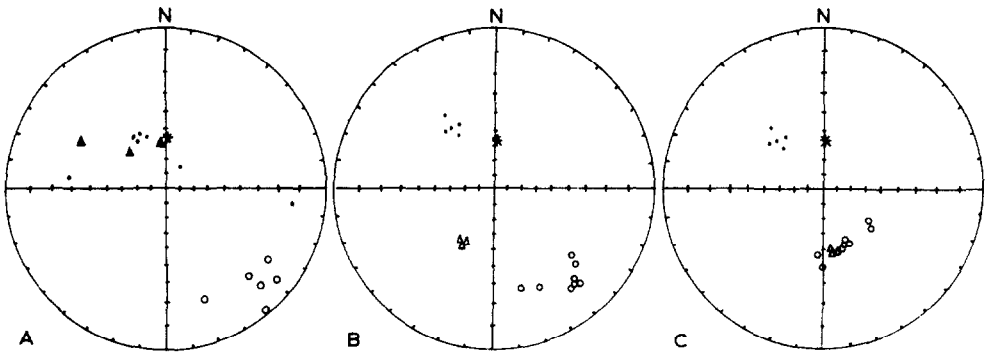


Fig.14. Eocene trachytes (triangles) and red beds (circles or dots), from the Tunceli-region. Stereographic projections of the directions of magnetization: A. total N.R.M.; B. characteristic components after demagnetization; C. characteristic components after demagnetization and dip-correction. Further explanation see Fig.6.

eliminated in a.c. magnetic fields between 200 and 400 Oe (peak value) and another "reversed" component appeared with a direction common to the directions of the sediment samples, being thus characteristic of the Eocene l local geomagnetic field.

The intensity of the siltstones varied between $2 \cdot 10^{-6}$ and $15 \cdot 10^{-6}$ e.m.u./cm^3, and their Q-values between 0.2 and 2.6. A.c. demagnetization also revealed a secondary component (Fig.13B) which in most cases could be eliminated in the a.c. magnetic field trajectory between O and 2,000 Oe. The directions of the remaining hard magnetizations have been plotted in Fig.14B, being characteristic for all samples. From some samples the secondary components could not be eliminated with 3,000 Oe; consequently, the direction of their original magnetization could not be determined, and these samples have been discarded from computation (Site E: samples TUN 20-22; samples TUN 8, 11, 12).

Considering the directions after unfolding has been applied, we see a normal and a reversed group (Fig.14C). An average direction has been computed for 16 samples with $D = 155°$, $I = -48°$ and $\alpha_{95} = 6°$. Giving unit weight to sites (N=4), the average direction becomes $D = 152{,}5°$, $I = -48°$ and $\alpha_{95} = 16°$.

DISCUSSION OF RESULTS

A review of the Jurassic and Cretaceous data obtained from Europe, Africa, the Arabian Shield and Turkey is given in Table III, and in Fig.15 the virtual pole positions are plotted.

The positions of the Jurassic, Cretaceous and Eocene virtual poles from Turkey differ notably from the Eurasian data (Fig.15, full symbols) obtained so far from stable parts of this area. A similar deviation is displayed by the Jurassic and Cretaceous pole positions from Lebanon and Israel (Van Dongen et al., 1967; Nur and Helsley, 1967).

It might be possible that since the Jurassic period, local structural disturbances have affected our regions. The regions sampled lie to a maximum distance of 100 km only from the North Anatolian faultzone, which is well known to be a very active earthquake belt. Pavoni (1961) considers this zone as an active dextral strike-slip fault at least since the Jurassic and we might expect to find dragging regions on either side. If we consider, however, the deviation between our Cretaceous data from the Pontide chains and the Eurasian data (Fig.15), it is obvious that a sinistral rotation since the Cretaceous period is needed to reconcile the Eurasian data and our results. Since a dextral faultzone, moreover, can be held responsible only for dextral dragging of the adjoining regions, our conclusion has to be that the deviation between our Turkish data and the Eurasian data can not be explained by local structural disturbances due to the North Anatolian faultzone.

On the other hand there seems to be a rather marked agreement between the Upper Mesozoic data from Turkey and Africa[1] and the Arabian

[1]While this paper was in preparation, new paleomagnetic studies yielded two new Cretaceous pole positions from rocks of Central and South Africa. Both of these are situated in northern Canada (at about 60°N, 100°W) as well (Briden, 1967; Jones et al., 1967).

TABLE III

Some ancient pole positions, derived from reliable paleomagnetic research on Jurassic, Cretaceous and Eocene rocks in Eurasia, Africa, the Arabian Shield and Turkey[1]

Nr. in Irving (1964)	Nr. in Fig.15	Continental unit and locality	Age	Pole position	Number of samples and sites	α_{95}	Author and year
10.05	10.05	Eurasia, U.S.S.R.	Cretaceous (Albian)	62°N, 169° E.	–	–	Khramov (1958)
10.06	10.06	Eurasia, U.S.S.R.	Cretaceous	60°N, 165° E.	–	–	Khramov (1958)
10.15	10.15	Africa, Mozambique (Lupata)	Cretaceous (109 m.y.)	62°N, 101° W.	61, 7 sites	3°	Gough and Opdyke (1963)
–	2	Arabian Shield, Lebanon	Cretaceous (Aptian)	38°N, 78°W.	15, 5 sites	5,5°	Van Dongen et al. (1967)
–	3	Arabian Shield, Israel	Upper Cretaceous	42°N, 96°W.	–	–	Nur and Helsley (1967)
–	4	Arabian Shield, Israel	Lower Cretaceous	60°N, 106°W.	–	–	Nur and Helsley (1967)
–	5	Turkey (Niksar)	Cretaceous	51°N, 64°W.	9, 2 sites	16°	present paper
–	6	Turkey (Gümüshane)	Cretaceous	61°N, 82°W.	31, 6 sites	5,5°	present paper
–	7	Turkey (Gümüshane)	Upper Cretaceous–Eocene	69°N, 99°W.	23, 6 sites	3,5°	present paper
–	8	Turkey (Tunceli)	Eocene	65°N, 66°W.	16, 4 sites	6°	present paper
9.51	9.51	Africa, Rhodesia and South Africa (Karroo + Stormberg)	Upper Triassic–Lower Jurassic	69°N, 97°W.	mean pole for various papers on the Karroo- and Stormberg-system investigations		
–	9	Arabian Shield, Lebanon	Upper Jurassic (Kimmeridgian)	1°S, 60°W.	20, 6 sites	3°	Van Dongen et al. (1967)
–	10	Turkey (Bayburt)[2]	Jurassic	(39°N, 95°W)	8, 1 site	8°	present paper

[1]The pole positions mentioned here, are those so far determined from samples subjected to extensive stability tests, such as a.c. or thermal demagnetization techniques. The semi-angle of the 95% confidence (α_{95}) is given in respect to the computation of the average, giving unit weight to samples.

[2]Entered in brackets, because of unreliability.

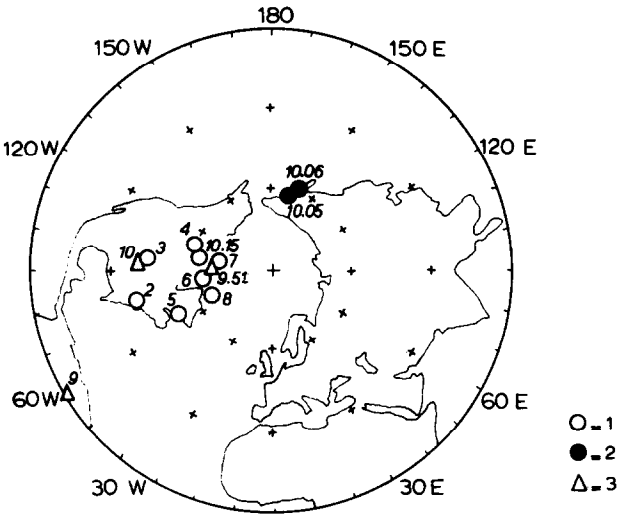


Fig.15. Some Mesozoic pole positions for Europe, Africa, the Arabian Shield and Turkey. The pole positions plotted here, are those so far determined from samples subjected to extensive stability tests, such as a.c. magnetic field or thermal demagnetization techniques. The numbers refer to Table III or Irving's list (1964). *1* = Cretaceous poles Africa and Near East; *2* = Cretaceous poles Eurasia; *3* = Jurassic poles Africa and Near East.

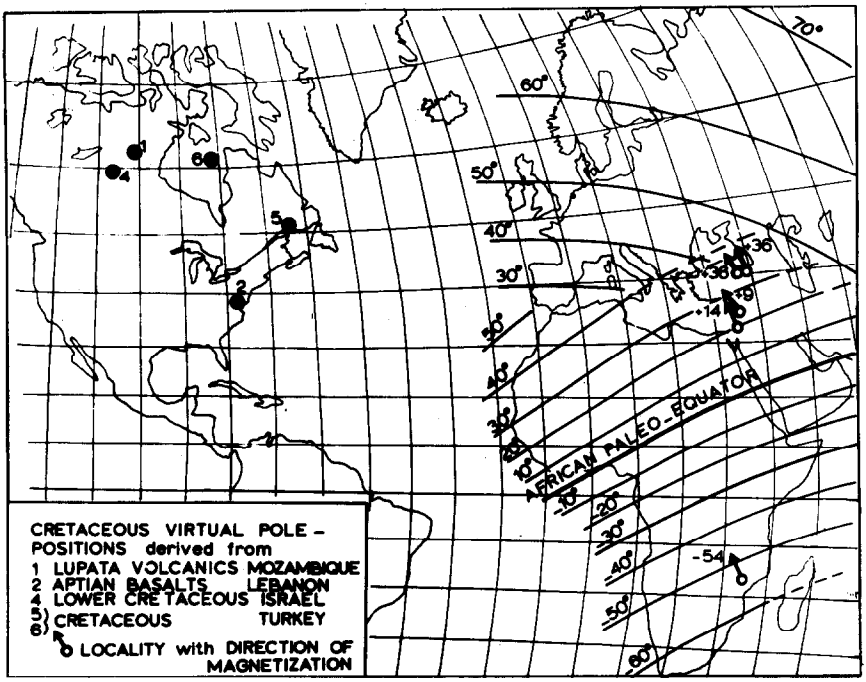


Fig.16. Pole positions from Africa, the Arabian Shield and Turkey for the Cretaceous period. Paleo-isoclines have been drawn for most reliable virtual pole positions from Africa (Lupata Alkine volcanics) and Europe (U.S.S.R.). Sampling localities are indicated with direction of magnetization; the numbers of the pole positions refer to Table III.

Shield (Fig.15, open symbols). It should also be remembered, that a Permian virtual pole position from Amasra, Turkey, also is situated in "the zone, covered by the Permian poles thus far determined for the African continent (Nairn, 1959, 1964; Opdyke, 1964)", as has been stated by Gregor and Zijderveld (1964). The relationship between our data and the results from Africa and Eurasia is once more demonstrated by Fig.16: paleo-isoclines have been constructed for the mean Cretaceous virtual pole positions from the different continental units, viz., Eurasia and Africa.

The picture shows convincingly that the magnetizations measured in Turkey and the Arabian Shield favour a common Cretaceous virtual pole position for Africa, the Arabian Shield and Turkey. The azimuthal directions of magnetization from Turkey, plotted where their rocks were sampled, are more or less perpendicular to the paleo-isoclines for Africa, and the values of their inclinations come very close to that of the nearby isoclines.

It can be argued that the inclinations from Turkey fit into the European isocline-pattern as well, both for the Permian from Amasra (Gregor and Zijderveld, 1964) as for the Cretaceous (present paper). The declinations, however, deviate largely from the directions to be expected locally for the mean European Permian and Cretaceous poles. If we try to explain these deviations by a rotation of Turkey we need an 80° counter-clockwise rotation for the Permian and a 50° clockwise rotation for the Cretaceous to reconcile the European and Turkish results. This would imply a complicated and somewhat preposterous geotectonic history.

It is hoped that further work on the Permian and Mesozoic of the Mediterranean will give more details about the earlier geological history of this part of the Tethys zone.

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